Practice problems #13

1. Let W be a subspace of $\mathbf{R^n}$. Show that $W^\perp$ is also a subspace of $\mathbf{R^n}$.

2. True or False. Justify:
   If u is in both W and $W^\perp$, then $u = 0$.

Answers or Hints on page 2.

#1: Follow the outline of the proof given in class.

#2: True. (Hint: If $u \neq 0$ is in $W \cap W^{\perp}$, then what should happen?)